## Assignment-1 (Digital Logic Design)

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| **Q.1** | **Do as directed:** | |
| | **1.** | $(516)_7$ = (                    $)_{10}$= (                $)_{16}$ |
| | **2.** | $(250.5)_{10}$ = (                $)_8$= (              $)_4$ |
| | **3.** | $(2ED)_{16}$=(              $)_8$=(                      $)_2$ |
| | **4.** | $(38)_9$=(              $)_5$=(                      $)_2$ |
| | **5.** | Obtain the 9's and 10's complement of $(864)_{10}$. |
| **Q.2** | **Do as directed:** | |
| | **1.** | $(198)_{12}+(12121)_3$=(                $)_8$ |
| | **2.** | Determine the value of base x if $(50)_x=(203)_4$ |
| | **3.** | Given the two binary numbers X = 1010101 and Y = 1001011, perform the subtraction X-Y using 1's complements. |
| | **4.** | Using 10's complement perform $(4572)_{10}-(2102)_{10}$. |
| | **5.** | Multiply the $(135)_6$ and $(43)_6$ in the given base without converting to decimal. |
| **Q.3** | **Do as directed:** | |
| | **1.** | $(347)_{10}$ = (                $)_2$= (              $)_8$= (            $)_5$ = (              $)_{16}$ = (            $)_{BCD}$ |
| | **2.** | $(11010111.110)_2$ = (                $)_{10}$= (                  $)_{12}$= (            $)_{16}$ |
| | **3.** | Obtain the 9's and 10's complement of $(389.61)_{10}$. |
| **Q-4** | **Do as directed:** | |
| | **1.** | Multiply the $(267)_8$ and $(71)_8$ in the given base without converting to decimal. |
| | **2.** | $(103)_4 + (50)_7$ = (              $)_9$ |
| | **3.** | Determine the value of base b if $(211)_b=(152)_8$ |
| | **4.** | Given the two binary numbers X = 11010 and Y = 1101, perform the subtraction X-Y using 2's complement. |
| | **5.** | Using 9's complement perform $(582)_{10}-(1002)_{10}$. |
| **Q.5** | **(a)** | Define duality principle and explain it with the help of example. Find the complements of the functions **F1 = x'yz' + x'y'z** and **F2 = x(y'z'+yz)** by taking their duals and complementing each literal. |
| | **(b)** | Demonstrate by means of truth tables the validity of the De Morgan's theorems for three variables. Find the complement of **F = a(b'c'+bc)** by applying De Morgan's theorem as many times as necessary. |
| **Q.6** | **(a)** | Demonstrate by means of truth table the validity of the distributive law of + over $\cdot$. Also show that the NOR and NAND operators are not associative. |
| | **(b)** | Prove that a positive-logic AND gate is a negative-logic OR gate and vice-versa. |
| **Q.7** | **(a)** | Express the Boolean function **F(p, q, r, s)=s(p'+q)+q's** in a sum of minterms and a product of maxterms. |
| | **(b)** | Given the Boolean function: **F=xy+x'y'+y'z** |
| | | **1.** Implement it with *only* OR and NOT gates. |
| | | **2.** Implement it with *only* AND and NOT gates. |
| **Q.8** | **(a)** | What is the difference between canonical form and standard form? Express the Boolean function **F(p, q, r)=(pq+r)(q+pr)** in a sum of minterms and a product of maxterms. |
| | **(b)** | Realize 2 input X-OR gate using NOR gates only. |
| **Q.9** | **(a)** | Prove that: |
| | | **1.** wx+x'y+wy=wx+x'y |
| | | **2.** (AB+C+D)(C'+D)(C'+D+E)=ABC'+D |
| | | **3.** (A+B)'(A'+B')'=0 |
| **Q.10** | **(a)** | Simplify the following Boolean expressions. |
| | | **1.** F(w,x,y,z) = xy+wy'+wx+xyz |
| | | **2.** F(p, q, r, s) = (p'+q) (p+q+s)s' |

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| | **(b)** | Simplify the following Boolean functions using Karnaugh map: | | |
| | | **1.** | $F(A,B,C,D)=\Pi(0,1,2,3,4,10,11)$ | |
| | | **2.** | $F(w,x,y,z)=\Sigma m(0,1,2,4,5,12,13,14)$ + don't care conditions $\Sigma d(6,8,9)$. | |
| **Q.11** | Implement Boolean functions | | | |
| | **(a)** | F= (A +B') (CD+E) using only NAND gates. | | |
| | **(b)** | F=A (B+CD) +BC' with only NOR gates. | | |
| | **(c)** | F=x'y+xy' using only four NAND gates. | | |
| **Q.12** | Simplify the function $F(w,x,y,z)=\Sigma(0,1,2,8,10,11,14,15)$ using tabulation method. | | | |
| **Q.13** | Differentiate between combinational logic circuit and sequential logic circuit. Design a combinational circuit that accepts a three-bit number and generates an output binary number equal to the square of the input number. | | | |
| **Q.14** | Design a combinational circuit that multiplies by 5 an input decimal digit represented in BCD. The output is also in BCD. Show that the output can be obtained from the input lines without using any logic gates. | | | |
| **Q.15** | Design a combinational circuit that converts a four-bit reflected-code number to a four-bit binary number. Implement the circuit with exclusive-OR gates. | | | |